

Project Overview:

AssetTrack is a web-based application designed to automate the management, tracking, and reporting of hardware assets within a software development team. The system will replace the current manual process of updating hardware records with an efficient, user-friendly interface, ensuring real-time accuracy and availability of asset information.

Functional Requirements:

1. User Management:

- **1.1 User Authentication:**
 - Users must log in with a secure username and password.
 - Support for Single Sign-On (SSO) and multi-factor authentication.
- **1.2 User Roles:**
 - Different roles with varying levels of access (e.g., Admin, Manager, Developer).
 - Admins can manage roles and permissions.

2. Asset Management:

- **2.1 Asset Registration:**
 - Ability to add new assets (laptops, screens, accessories) to the system.
 - Input fields for asset details: type, brand, model, serial number, purchase date, warranty expiration, etc.
- **2.2 Asset Allocation:**
 - Assign assets to team members, tracking both current and previous ownership.
 - Track the history of asset transfers between users.
- **2.3 Expiration Tracking:**
 - Automatic notifications for upcoming warranty expiration dates.
 - Flag expired laptops and suggest actions (e.g., reassign as spares, decommission).

3. Automated Updates:

- **3.1 Inventory Sync:**
 - Automatic synchronization of asset status changes (e.g., updates when an asset is reassigned).
- **3.2 Asset Condition Reporting:**
 - Users can report the condition or issues with their assigned assets.
 - Option for automatic condition checks through system integrations (e.g., battery health for laptops).

4. Reporting and Analytics:

- **4.1 Inventory Dashboard:**
 - Overview of current assets, categorized by type, status, and user.
 - Graphical representation of asset allocation and status.

- **4.2 Usage Statistics:**
 - Detailed reports on asset usage, allocation history, and condition. (plus)
- **4.3 Alerts and Notifications:**
 - Customizable alerts for upcoming warranty expirations, low stock levels for accessories, etc.
 - Email and in-app notification support.

5. Search and Retrieval:

- **5.1 Advanced Search:**
 - Search functionality to find assets by serial number, user, status, type, etc.
- **5.2 Asset Retrieval:**
 - Ability to quickly retrieve and display detailed asset information.
 - Ability to get instant spare laptop with owner details

Non-Functional Requirements:

1. Performance/Scalability:

- Number of user will not be much ~100 and it is not expected to have load on the system

2. Security:

- The server is expected to be internal and all users are internal so no security requirements needed at this point other than authentication

3. Maintainability:

- The codebase should be modular and well-documented to facilitate maintenance and future enhancements.
- Regular updates and patches should be manageable with minimal disruption.

Technology Stack:

- **Left to the group**

Deliverables:

- Functional and non-functional requirements document
- System architecture and design diagrams
- Working prototype for initial feedback (with the supervisor)
- Fully functional web-based AssetTrack system